Tecnologico de Monterrey
Interpartial Exam 2 A
2026-Jan-May

Grade

Course: Calculus I

ID-number: ______________

Professor: Fco. Javier Vazquez Tavares

Date: 09/02/2026

Last Names and Names: ____

Group: 201

"*In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honestly and not to use unauthorized or illegal means to do so.*"

Signature: ______________

I Limits. Properties and Algebraically.

1. (6 **points**) Compute the limit of the function $f(x) = \frac{x^2-4}{x-2}$ as $x \to 2$.

2. (7 **points**) Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \to \infty$.

3. (7 **points**) Compute the limit of the function $f(x) = \frac{\sqrt{x+4}-5}{x}$ as $x \to 0$.

II Derivative using the power rule and properties of derivatives.

1. (10 **points**) Using the power rule, compute the derivative of the function $f(x) = \dfrac{3}{2x}$.

2. (10 **points**) Using the power rule, compute the derivative of the function $g(x) = 4\left[\left(\frac{x}{2}\right)^{12} - \left(\frac{x}{2}\right)^{6}\right]$.

III Derivatives of trascendental functions.

1. (6 **points**) Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = \sin(x) + \cos(x)$.

2. (6 **points**) Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = e^x + \ln(x)$.

3. (8 **points**) Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = \ln(5x)$.

IV Derivative by definition. $\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$.

1. (20 **points**) Compute the derivative of the function $f(x) = \sqrt{x}$ using the definition of derivatives.

2. (20 **points**) Compute the derivative of the function $f(x) = |x|$ at $x = 0$ using the definition of derivatives.